



MedCertify

Train for Healthcare & IT Careers

COMPTIA NETWORK+

CERTIFICATION

SYLLABUS

WWW.MEDCERTIFY.COM

ONLINE DELIVERY

100 INSTRUCTIONAL
CLOCK HOURS
10 WEEKS



OVERVIEW:

The CompTIA Network+ course prepares students for the CompTIA Network+ certification exam and a career as a Network Administrator. Students develop a full understanding of networking concepts by learning both the theory and technical skills needed for the role. The course covers essential topics such as network design, implementation, maintenance, troubleshooting, and security, helping students build a strong foundation for managing and supporting network environments.

Students will:

- Understand networking concepts, standards, and protocols.
- Learn about network devices, architectures, and configurations.
- Explore wide area networking (WAN) and remote access solutions.
- Develop skills in network security, monitoring, and maintenance.
- Troubleshoot, diagnose, and resolve network issues.
- Apply best practices for network management and performance optimization.



PROGRAM FEATURES

- ✓ Expert Guidance Provided Through Live Mentoring Support Sessions
- ✓ 6-Month Access to On-Demand Video Lessons Taught by Industry Leaders
- ✓ Flexible, Self-Paced Learning that Adapts to Your Schedule
- ✓ Dedicated Academic Coaches
- ✓ National Certification Exam Preparation Support



PROGRAM COURSES

- ✓ Workplace OSHA Training
- ✓ Career Development
- ✓ Professional Office Development
- ✓ CompTIA Network+ w/ Labs

PROGRAM OUTLINE

WEEKS

1-2

Networking Concepts

- Introduction to Network Communications
- LAB - Networking Topologies and Architecture
- Examining Network Standardization
- Introduction to TCP/IP and the OSI Model
- LAB - Introduction to the OSI Model
- Comparing TCP/IP to the OSI Model
- Encapsulation and Decapsulation
- Examining Network Topologies and Traffic Flows
- Networking Protocols and Services
- LAB - Networking Ports and Protocols
- Internet Protocol Number and Traffic Types
- Examining Public and Private Addressing
- LAB - IPv4 Network Addressing
- Examining IPv4 Address Classes
- Examining IPv4 Addressing
- Introduction to Subnetting
- Introduction to VLSM
- Introduction to CIDR
- Implementing IP Addressing
- LAB - Implementing IPv4 Network Services
- Examining Network Switches
- LAB - Switching Concepts
- Examining Network Routers
- LAB - Routing Concepts
- Examining Wireless Appliances
- Examining Firewalls and IPSs
- Examining Load Balancers and Proxies
- LAB - Routing Concepts
- Examining Wireless Appliances
- Examining Firewalls and IPSs
- Examining Load Balancers and Proxies
- LAB - Disaster Recovery Concepts
- Examining Network Storage
- Examining VPNs and QoS
- Examining the 802.11 Standard
- Examining the 802.3 Standard
- Examining Transceivers and Connector Types
- Examining Wireless Communications
- Examining Cloud Concepts
- Examining Cloud Components
- LAB - Cloud Networking Concepts
- Examining SDN and SD-WAN
- LAB - Software Defined Networking Concepts
- Examining VXLAN
- Examining Zero Trust Architecture

- Examining Infrastructure as Code
- Examining IPv6 Addresses
- Networking Concepts Review Quiz

WEEKS

3-4

Network Implementation

- Static Routing
- OSPF
- EIGRP
- BGP
- Route Selection
- Address Translation
- First Hop Redundancy Protocols
- Subinterfaces
- VLANs
- Interface Configuration
- Spanning Tree Protocol
- MTU
- Frequencies and Channels
- SSIDs
- Network Types
- Encryption
- Authentication
- Guest Networks
- Antennas
- Access Points
- Installation Implications
- Power
- Environmental Factors
- Network Implementation Review Quiz

WEEKS

5-6

Network Operations

- Common Documentation
- CLife-cycle Management
- CConfiguration/Change Management
- CSNMP
- CFlow Data and Packet Captures
- CLAB - Network Monitoring Concepts
- CBaseline Metrics
- CLog Aggregation
- CAPI Integration
- CPort Mirroring
- CNetwork Monitoring Solutions
- CDR Metrics

PROGRAM OUTLINE

- CDR Sites
- CHA Approaches-DR Testing
- CDHCP
- CDNS
- CTime Protocols
- CVPNs
- CLAB - Networking Appliances and Functionality
- CConnection Methods
- CLAB - Network Access and Management
- CJump Boxes
- CIn-Band vs Out-of-Band Management
- CNetwork Operations Review Quiz

WEEK 7

Network Security Fundamentals

- Data Security Concepts
- LAB - Network Security Concepts
- Examining Digital Certificates
- Examining Public Key Infrastructure
- Examining Identity and Access Management
- Examining Authentication Factors
- Examining Single Sign On
- Authentication with LDAP
- Examining AAA
- TACACS
- Time-Based Auth and Geofencing
- The Principle of Least Privilege
- Examining Authorization and Access Control Models
- Examining Physical Security
- Examining Deception Technologies
- Risks, Vulnerabilities, and Threats
- Examining the Principles of Security
- Regulatory Compliance and Audits
- IoT and SCADA
- Network Segmentation
- LAB - Network Hardening Techniques
- Network-Based Threats
- LAB - General Network Attacks
- Indicators of Network Attacks
- Malware and Social Engineering
- Network Security Defenses
- Network Security Fundamentals Review Quiz

WEEK 8-9

Network Troubleshooting

- Implementing a Troubleshooting Methodology
- Identifying the Problem in Troubleshooting
- Establishing and Testing Theories in Troubleshooting
- Establishing an Action Plan in Troubleshooting
- Implementing a Solution in Troubleshooting
- Verification and Documentation in Troubleshooting
- Examining Network Cabling Issues
- Examining Physical Interface Issues
- Examining Switching and Routing Issue
- Examining IP Addressing Issues
- Examining Network Traffic Issues
- Examining Wireless Network Issues
- Examining Network Commands in Windows
- LAB - Network Troubleshooting Tools and Techniques
- Examining Network Commands in Linux
- Examining Hardware-based Tools
- Examining Software-Based Tools
- Examining Basic Network Device Commands
- Network Troubleshooting Review Quiz

WEEK 10

FINAL EXAM

- CompTIA Network+ (N10-009) Final Exam



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